

3 Otology and Neurotology

Section Editor

Christine T. Dinh

Contributors

Christine T. Dinh

David Goldenberg

Bradley J. Goldstein

Jon E. Isaacson

Gregory T. Lesnik

Adam J. Levy

Elias M. Michaelides

Kari Morgenstein

Stuart A. Ort

Daniel I. Plosky

Julie A. Rhoades

3.0 Embryology and Anatomy of the Ear

◆ Embryology

Auricle

Week 6: Hillocks of His form from condensations of the first and second branchial arches. The first three hillocks are attributed to the first arch, and the second three hillocks are attributed to the second arch (**Table 3.1**).

Week 20: Adult configuration is achieved.

The auricle continues to grow and reaches 85% of adult size by 5 years of age.

External Auditory Canal

Week 8: Ectoderm of the first pharyngeal groove (branchial cleft) invaginates. Epithelial cells then grow as a solid core (meatal plug) toward the middle ear. The core begins to dissolve at ~21 weeks to create the external auditory canal (EAC). The lateral epithelium forms the skin of the bony EAC, while the medial epithelium forms the lateral surface of the tympanic cavity—particularly, the lateral layer of the tympanic membrane (TM). The EAC may not reach final adult size and shape until the early teenage years.

Tympanic Membrane

The tympanic membrane (TM) forms as the **ectoderm** of the first pharyngeal groove (EAC) medializes, thins the **mesenchyme** of the first branchial arch (fibrous middle layer of the TM or the “tympanic ring”), and abuts the **endoderm** of the first pharyngeal pouch (tubotympanic recess or middle ear). The **tympanic notch** (notch of Rivinus) lacks the fibrous middle layer.

Middle Ear

Week 3: Endoderm from the first pharyngeal pouch forms the tubotympanic recess. Pneumatization begins at 10 weeks to form the middle ear and auditory tube (eustachian tube, ET).

Week 4: Mesenchyme from the first and second pharyngeal arches fuses and begins to form the malleus and incus. The first **pharyngeal arch cartilage** is also referred to as Meckel’s cartilage, and the **second pharyngeal arch cartilage** is also referred to as Reichert’s cartilage. The cartilaginous ossicles

Table 3.1 Embryology of the auricle

Hillock	Arch	Auricular structure
1	1	Tragus
2	1	Crus of helix
3	1	Helix
4	2	Crus of antihelix
5	2	Antihelix
6	2	Antitragus and lobule

Note: There remains some controversy regarding the final contributions of hillocks 4, 5, and 6.

Table 3.2 Embryology of the Meckel and Reichert cartilages

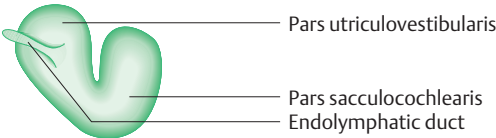
Cartilage	Arch	Ossicular structure
Meckel	1	Head and neck of malleus, body and short process of incus
Reichert	2	Manubrium of malleus, long process of incus, stapes suprastructure

Note: The manubrium of the malleus never completely ossifies.

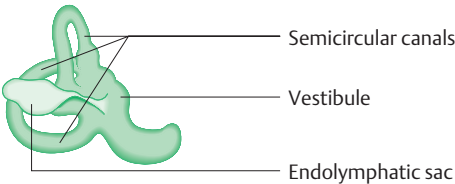
a Fourth week of development



b Fifth week of development



c Sixth week of development



d Eighth week of development

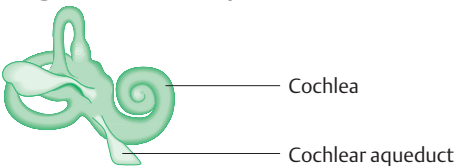


Fig. 3.1 Embryology of the inner ear. The otocyst forms from an epithelial thickening between the cutaneous ectoderm and neural groove in the third and fourth weeks of embryonic development. **(a)** This thickening invaginates and closes off to form a separate vesicle. **(b)** In the fifth week, the otocyst becomes infolded, forming the upper pars utriculovestibularis and the lower pars sacculocochlearis. **(c)** In the sixth week, the three semicircular canals form from the pars utriculovestibularis. **(d)** In the seventh to ninth weeks, the cochlear duct forms as a tubular extension of the pars sacculocochlearis and becomes coiled. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:158.)

form at adult size and shape by week 16. Subsequently, ossification occurs by endochondral bone formation (**Table 3.2**). The second pharyngeal arch cartilage also goes on to form the stapes blastema between weeks 4 and 5, which gives rise to the stapes suprastructure. The footplate is of otic capsule origin.

In general, the second pharyngeal arch cartilage forms the ossicular portions seen through the TM on otoscopy (manubrium of the malleus, long process of the incus, stapes superstructure). The first pharyngeal arch cartilage forms the ossicular portions superior to the TM (head of the malleus, body and short process of the incus). In addition, the tensor tympani is formed from the first pharyngeal arch, while the stapedial muscle is formed from the second pharyngeal arch.

Inner Ear

Week 3: The otic placode forms from ectoderm of the first pharyngeal groove. It invaginates and is completely encircled in mesoderm and termed the otocyst by week 4. The pars superior (semicircular canals and utricle) develops prior to the pars inferior (sacculle and cochlea). The membranous labyrinth is complete by week 15 or 16. Ossification occurs between weeks 20 and 25 (**Fig. 3.1**).

♦ Anatomy

Auricle

The auricle and the external auditory canal are referred to as the external ear. The auricle is made of elastic cartilage with perichondrium and skin that is tightly adherent anteriorly and loosely adherent posteriorly (**Fig. 3.2**). It is attached to the head by the extension of cartilage into the ear canal and anterior, superior, and posterior ligaments and muscles.

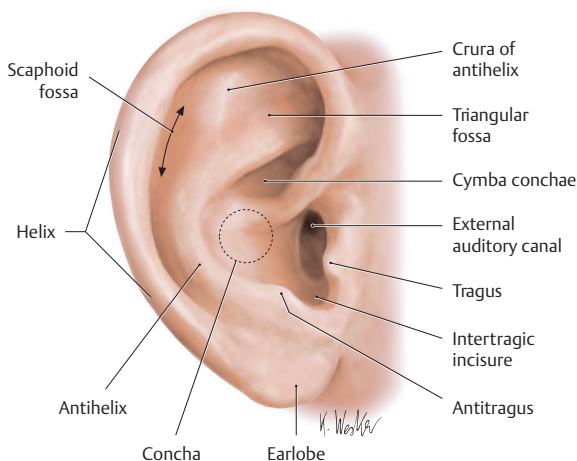


Fig. 3.2 Anatomy of the auricle. (From Gilroy AM, MacPherson BR, Ross LM. *Thieme Atlas of Anatomy, Head and Neuroanatomy*. 1st ed. Stuttgart/New York: Thieme; 2007. Illustration by Karl Wesker.)

External Auditory Canal

The EAC is one-third cartilaginous (lateral) and two-thirds osseous (medial). The EAC contains fibrocartilage, not elastic cartilage. The skin lining the cartilaginous EAC is thick with hair follicles and sebaceous and ceruminous glands. The medial skin is thin without any adnexal structures or subcutaneous tissue, is extremely pain-sensitive, and is the only site in the entire body with skin lying directly on periosteum. The EAC is innervated by contributions from the trigeminal nerve (cranial nerve [CN] V₃, auriculotemporal branch), the cervical plexus (C3, great auricular nerve), the vagus nerve (CN X, Arnold's nerve), and the facial nerve (CN VII). C2 and C3 contributions from the lesser occipital nerve innervate the inferior auricle, but not the EAC.

Arnold's reflex occurs when manipulation of the ear canal produces coughing. **Hitzelberger sign** is numbness over the posterior aspect of the EAC due to impaired facial nerve function from a cerebellopontine angle tumor. The **fissures of Santorini** are two vertical fissures in the anteroinferior aspect of the cartilaginous EAC, while the **foramen of Huschke** (foramen tympanicum) is a developmental defect in the anteroinferior aspect of the bony EAC. Infection and malignancies can travel through these channels to the temporomandibular joint, parotid gland, and infratemporal fossa.

Tympanic Membrane

The TM is made up of three layers. From lateral to medial, they are the squamous epithelium, a radiating and circular fibrous layer, and an inner mucosal layer. The TM is ~10 mm in height and 9 mm in width. Circumferentially, the fibrous anulus sits within the bony anulus but is discontinuous superiorly at the **tympanic notch**. Inferiorly to the tympanic notch, the TM has a well-organized fibrous middle layer and is known as the **pars tensa**. At the level of the tympanic notch and above, the middle fibrous layer is less organized and is known as the **pars flaccida**. **Prussak's space** is at the level of the pars flaccida, bounded laterally by the TM and medially by the neck of the malleus. It is a frequent site of retraction and cholesteatoma formation.

Middle Ear

The middle ear is divided into three pneumatized spaces. The epitympanum is the middle ear space above the TM, the mesotympanum is the middle ear space medial to the TM, and the hypotympanum is the middle ear space below the TM. The protympanum is the anterior portion of the mesotympanum, where the entrance of the ET is located. The boundaries of the middle ear space are:

- Lateral: tympanic membrane
- Superior: tegmen
- Inferior: jugular bulb
- Anterior: carotid canal, ET
- Posterior: mastoid via facial recess or retrofacial cells
- Medial: cochlear promontory and labyrinthine wall

Ossicles

- **Malleus.** The **malleus** (hammer) consists of a manubrium (handle), anterior and lateral processes, neck, and head. It is incorporated into the TM from the lateral process to the tip of the manubrium (at the umbo of the TM). The tensor tendon runs from the cochleariform process of the middle ear to the medial surface of the neck and manubrium of the malleus.
- **Incus.** The **incus** (anvil) is the largest ossicle and articulates with the malleus head in the epitympanum. It has a body and short, long, and lenticular processes. The short process is held in place by the posterior incudal ligament. Both the incudomalleolar and incudostapedial joints are synovial diarthrodial joints.
- **Stapes.** The **stapes** (stirrup) is the smallest ossicle and articulates with the lenticular process of the incus. It has a capitulum, an anterior and a posterior crus, and a footplate. The stapedial tendon runs from the pyramidal eminence to the posterior surface of the capitulum or the posterior crus. The footplate is kept in position by the annular ligament.

See **Fig. 3.3**. *Note:* The tensor tympani muscle is innervated by CN V₃, while the stapedius muscle is innervated by CN VII.

Eustachian Tube

The ET measures ~18 mm at birth and 35 mm in adulthood. The lateral one-third is bony, is located at the anterior mesotympanum (protympanum), and is lined with ciliated cuboidal epithelium. The medial two-thirds is cartilaginous, terminates at the torus tubarius of the nasopharynx, and is lined with pseudostratified columnar epithelium. The ET

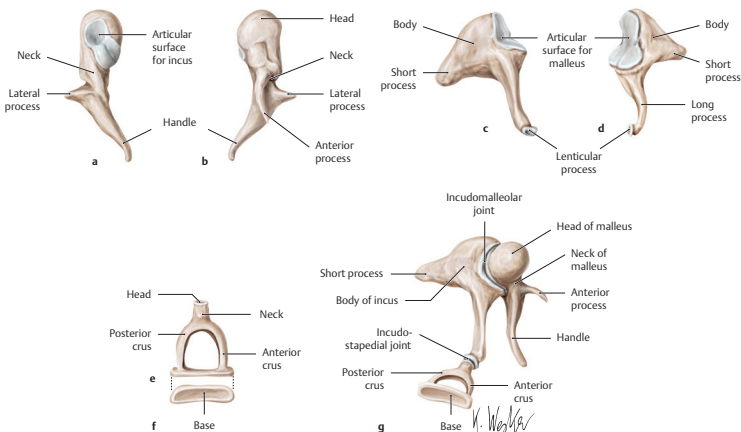


Fig. 3.3 Middle ear ossicles. Malleus, posterior (a) and anterior (b) views. Incus, medial (c) and anterolateral (d) views. (e) Stapes, superior and medial views. (f) Medial view of the ossicular chain. (From Gilroy AM, MacPherson BR, Ross LM. *Thieme Atlas of Anatomy, Head and Neuroanatomy*. 1st ed. Stuttgart/New York: Thieme; 2007. Illustration by Karl Wesker.)

functions to ventilate, clear secretions from, and protect the middle ear. It is closed at rest, and its primary dilator is the tensor veli palatini muscle (CN V₃).

Inner Ear

Bony Labyrinth

The cochlea is the auditory portion of the inner ear. It is shaped like a spiral around its central axis, called the modiolus. It has $2\frac{1}{2}$ turns and is divided in the basal, middle, and apical turns. The perilymphatic fluid of the basal turn of the cochlea communicates with the subarachnoid space through the **cochlear aqueduct** (a bony channel that houses the periotic duct).

The vestibular system consists of the otolith organs of the vestibule (utricle and saccule), the endolymphatic sac, and the semicircular canals (SCCs). The vestibule is the central chamber of the vestibular portion of the inner ear. Its medial wall has two depressions. The **elliptical recess** is located posteriosuperior and contains the macula of the utricle, while the **spherical recess** is located anteroinferior and contains the macule of the saccule. The **vestibular aqueduct** is the bony channel that contains the **endolymphatic sac**; it communicates with the vestibule posteroinferiorly via the **endolymphatic duct**.

The SCCs are three paired canals that lie at $\sim 90^\circ$ angles to one another. The canals are referred to as the horizontal (or lateral) SCC, the superior (or anterior) SCC, and the posterior SCC. Each SCC has ampullated and nonampullated ends that communicate with the vestibule. All ends connect to the vestibule separately except the nonampullated ends of the posterior and superior SCCs, which fuse and join the vestibule together as the common crus. The horizontal canal lies $\sim 30^\circ$ from the true horizontal.

Membranous Labyrinth

The membranous labyrinth is the fluid-filled chamber within the bony labyrinth. The membranous labyrinth contains endolymph, which has similar consistency to intracellular fluid (high potassium, low sodium). The membranous labyrinth is surrounded by perilymph, which has similar consistency to extracellular fluid (high sodium, low potassium).

The cochlear duct (or scala media) is the membranous portion of the cochlea, contains endolymph, and houses the auditory hair cells of the spiral organ (organ of Corti; **Fig. 3.4**). The scala media is flanked by two fluid-filled spaces called the scala vestibuli and scala tympani. The perilymph of the scala vestibuli and scala tympani communicate at the helicotrema of the apex of the cochlea. The endolymphatic fluid of the scala media connects to the saccule through the ductus reuniens. It also connects to the subarachnoid space through the periotic duct of the cochlear aqueduct.

The utricle and the saccule are the membranous portions of the vestibule. They contain endolymph but are surrounded by perilymph. The utricle and

sacculi are otolith organs that contain gelatinous membranes coated with otoconia and sense linear acceleration. The utricle senses movement in the horizontal plane, and the saccule senses motion in the vertical plane as well as gravitational force.

The semicircular ducts are the membranous channels within the SCCs. The semicircular ducts contain endolymph and are surrounded by perilymph. Each duct has an ampullated end, which houses the crista ampullaris, a cone-shaped organ containing hair cells that detect angular acceleration.

The endolymphatic sac contains endolymph and is located in the vestibular aqueduct within the dura of the posterior fossa. The endolymphatic sac communicates with the saccule via the endolymphatic duct.

Internal Auditory Canal

The internal auditory canal (internal acoustic meatus) is the bony channel of the posterior fossa that houses the facial nerve (CN VII) and vestibulocochlear nerve (CN VIII). It begins medially at the porus acusticus internus and runs laterally to end at the fundus. At the fundus, it is split horizontally by the falciform or transverse crest. The superior portion of the canal above the falciform crest is then split again into an anterior and posterior compartment by the vertical crest or Bill's bar. In the fundus, the facial nerve is located anterosuperior, the cochlear nerve is located anteroinferior, the superior part of the vestibular nerve is located posterosuperior, and the inferior part of the vestibular nerve is located posteroinferior.

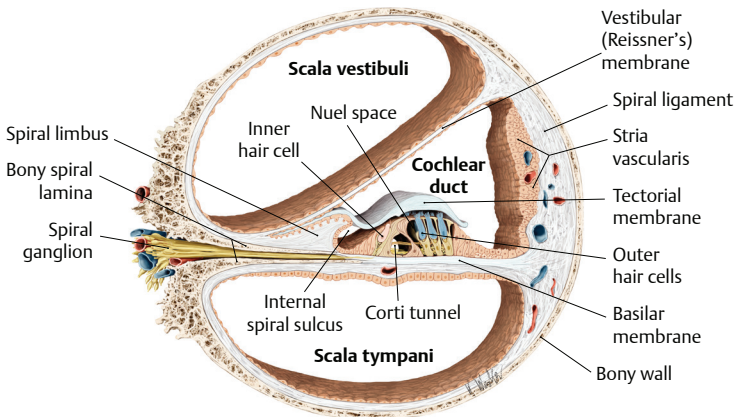


Fig. 3.4 The cochlear duct. (From Gilroy AM, MacPherson BR, Ross LM. *Thieme Atlas of Anatomy, Head and Neuroanatomy*. 1st ed. Stuttgart/New York: Thieme; 2007. Illustration by Karl Wesker.)

3.1 Otologic Emergencies

3.1.1 Sudden Hearing Loss

◆ Key Features

- Sudden hearing loss can be sensorineural or conductive in nature.
- Audiogram is performed to differentiate between sensorineural and conductive hearing loss.
- Idiopathic sudden sensorineural hearing loss is diagnosed when other causes have been excluded.
- Early treatment of idiopathic sudden sensorineural hearing loss with oral and/or intratympanic steroids is commonly recommended.

Sudden hearing loss is the rapid onset of subjective hearing impairment over 72 hours or less. Sudden sensorineural hearing loss (SSNHL) is one type of sudden hearing loss. The potential causes of SSNHL are broad: infection, vascular insult, trauma, autoimmune, neoplasm, acoustic trauma, drug-related, iatrogenic, and idiopathic. One common criterion used to define SSNHL is at least a 30-decibel (dB) reduction in sensorineural thresholds in at least three consecutive frequencies compared to baseline. Early implementation of high-dose systemic and/or intratympanic steroids is commonly recommended.

◆ Epidemiology

SSNHL affects 5 to 20 per 100,000 people in the United States. Men and women are affected equally. Most commonly it affects middle-aged people.

◆ Clinical

Signs and Symptoms

Patients may report sudden subjective hearing loss, ear fullness, and/or tinnitus in one or both ears. Dysequilibrium or vertigo can also occur. A history of upper respiratory tract infection, head trauma, cardiac surgery, ototoxic drug intake, otologic surgery, barotrauma, and loud noise exposure may precede onset of symptoms. Preexisting history or family history of autoimmune disorder may be ascertained.

Differential Diagnosis

Idiopathic SSNHL is diagnosed when other causes have been excluded. The potential causes of SSNHL are broad and include viral infections, syphilis, vascular insult, trauma, autoimmune disease, internal auditory canal or cerebellopontine angle neoplasm, acoustic trauma, ototoxic drugs, barotrauma, perilymphatic fistula, endolymphatic hydrops, Menière's disease,

and iatrogenic injury. Middle ear or external auditory causes such as middle ear effusion and cerumen impaction are causes of conductive hearing loss and should be excluded.

◆ Evaluation

Physical Exam

A complete head and neck examination, including cranial nerve assessment, should be performed. External and middle ear disorders should be assessed on otoscopy. Neurologic exam can identify signs of central or systemic disorders. Vestibular exam can demonstrate any associated vestibular weakness. A tuning fork and fistula test are important as well.

Audiometry

Pure tone and speech audiometry and tympanogram are required. Causes of conductive hearing loss can be excluded on these examinations. A poor word recognition score out of proportion to pure tone thresholds and rollover should raise suspicion for a retrocochlear lesion. A low-frequency sensorineural hearing loss is suspicious for endolymphatic hydrops or Menière's disease. A notch at 3 to 4 kHz suggests acoustic trauma. A commonly accepted definition for SSNHL is a reduction in sensorineural hearing of at least 30 dB in at least three contiguous frequencies compared to baseline; however, in clinical practice, SSNHL includes hearing loss of lesser degrees.

Imaging

Magnetic resonance imaging (MRI) of the brain with and without gadolinium enhancement is obtained to rule out retrocochlear or cerebellopontine angle tumor.

Intracranial disorders can also be excluded on MRI. Auditory brainstem response (ABR) testing and follow-up audiometry are acceptable alternatives to MRI if appropriate counseling regarding limitations is performed. In patients with a presenting history of head trauma, computed tomography (CT) of the temporal bone without contrast is considered to assess for temporal bone fractures and pneumolabyrinth causing SSNHL.

Labs

Routine ordering of laboratory tests in patients with unilateral idiopathic SSNHL is not recommended. In patients with fluctuating and bilateral SSNHL, additional testing is suggested: (1) rapid plasma reagin (RPR) test or a fluorescent treponemal antibody absorption (FTA-ABS) test to rule out syphilis, (2) erythrocyte sedimentation rate (ESR) to screen for inflammatory disorder, (3) rheumatoid factor, antinuclear antibody, and antineutrophil cytoplasmic antibodies to assess for autoimmune etiology, (4) Lyme serology if risk factors are present, (5) thyroid function tests, (6) and (7) glucose testing for diabetes.

◆ Treatment Options

Treatment is tailored to the etiology of SSNHL. However, approximately 85% of cases are considered idiopathic. SSNHL may resolve spontaneously in up to 65% of patients. Early treatment with oral and/or intratympanic steroids is commonly recommended as initial therapy for SSNHL, as prolonged delay in treatment may result in permanent hearing loss. Most clinicians prescribe oral prednisone (1 mg/kg daily) for 1 week with 1 week additional taper. Intratympanic dexamethasone has been used in various concentrations but usually requires at least three injections. Routine use of antivirals, thrombolytics, vasodilators, vasoactive substance, and antioxidants is not recommended. Hyperbaric oxygen therapy may be offered within three months of onset of idiopathic SSNHL.

◆ Outcome and Follow-Up

Although approximately two-thirds of patients with idiopathic SSNHL experience some recovery without treatment, early treatment is associated with better hearing outcomes. Follow-up audiograms to assess for recovery or progression are performed within 6 months of diagnosis and treatment of idiopathic SNHL. Hearing rehabilitation is offered to patients with incomplete recovery. Tinnitus associated with hearing impairment should also be addressed and treated appropriately.

3.1.2 Ear and Temporal Bone Trauma

◆ Key Features

- Auricular injuries include lacerations, avulsions, blunt injury with hematoma, thermal injury, and frostbite. Treatment depends on etiology.
- Penetrating and blunt trauma to the temporal bone can cause hearing loss, dizziness, facial paralysis, cerebrospinal fluid leak, and life-threatening hemorrhage. Early recognition and treatment of complications can reduce morbidity and mortality.

Penetrating and blunt trauma can affect all aspects of the ear and temporal bone, including the auricle, external auditory canal (EAC), tympanic membrane, ossicles, inner ear, carotid artery, venous sinuses, tegmen, and facial nerve. High suspicion and prompt recognition and treatment of associated temporal bone injuries is important in reducing trauma-related complications.

◆ Epidemiology

The most common cause of temporal bone fractures is motor vehicle accidents. Assault is the second most common cause; males are more commonly affected. Approximately 20% of skull fractures involve the temporal bone. Soft tissue injuries to the auricle are common.

◆ Clinical

Signs and Symptoms

Presentation depends upon the type of trauma and the location and extent of injury. Trauma to the auricle can be due to laceration, avulsion, blunt injury, thermal injury, or frostbite. Trauma to the temporal bone can be from acoustic trauma, barotrauma (e.g., diving), penetrating trauma (e.g., cotton tip applicator, gunshot wound), or blunt trauma (e.g., motor vehicle accident, assault). Thermal injury can also injure the tympanic membrane; for example, hot slag from a blast.

Auricular Injury

Blunt injury to the auricle may result in auricular pain, edema, erythema, and hematoma. Localized pain and bleeding are common after lacerations and avulsions of the auricle. Burn injuries can cause blistering, skin de-epithelialization, and cartilage exposure. The auricle is a common site of frostbite injury, which may present as a spectrum of findings over several weeks, from clear blistering, hemorrhagic blisters, or a dry insensate wound to blackened tissue demarcation and necrosis.

External, Middle, and Inner Ear Injuries

EAC trauma and traumatic tympanic membrane perforations can present with pain, bloody otorrhea, and conductive hearing loss. Injuries affecting the middle ear present as hemotympanum (blood in the middle ear) and/or conductive hearing loss and can involve ossicular fracture or discontinuity. Trauma involving the inner ear can cause sensorineural hearing loss (e.g., cochlear concussion, otic capsule violating fracture) and vertigo (e.g., benign paroxysmal positional vertigo, labyrinthine concussion, perilymphatic fistula, vestibular loss).

Other Injuries

Penetrating trauma and temporal bone fractures can involve the facial (fallopian) canal and cause immediate and delayed facial palsies. The perigeniculate area is the most common site of facial nerve injury in temporal bone fractures. Blunt trauma to the extratemporal facial nerve can also cause facial paralysis. Penetrating trauma and temporal bone fractures involving the middle cranial fossa plate, posterior cranial fossa plate, or otic capsule can cause leakage of cerebrospinal fluid (CSF) into the temporal bone, which can manifest as clear middle ear effusion, otorrhea, or rhinorrhea. Injuries to the horizontal and vertical petrous internal carotid artery, sigmoid sinus, or jugular bulb can cause life-threatening hemorrhage and hemodynamic instability.

Temporal Bone Fractures

Temporal bone fractures are most often seen in patients with severe head trauma. Additional injuries requiring resuscitation and multi-team care are common. ENT findings may include otorrhea, hearing loss, nystagmus, and facial paralysis. A bruise over the mastoid (**Battle sign**) is an indication of fracture of the middle cranial fossa and injury of the posterior auricular artery. Periorbital ecchymosis (**raccoon eyes**) suggests a basal skull fracture.

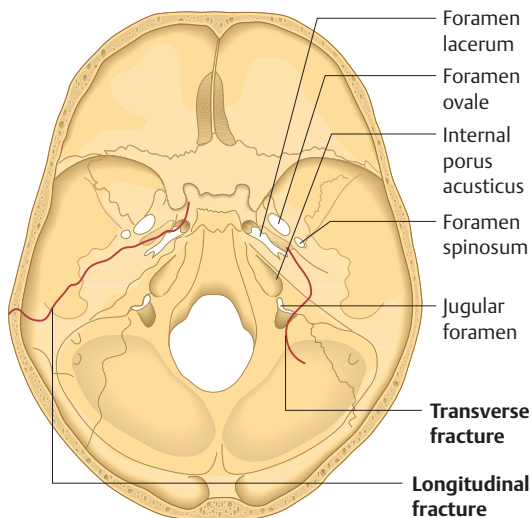


Fig. 3.5 Temporal bone fractures: a typical longitudinal temporal bone fracture (left) and transverse temporal bone fracture (right). (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:303.)

Temporal bone fractures are classified as either **otic-sparing or otic-violating fractures**. Otic-violating fractures are more commonly associated with sensorineural hearing loss. Older classifications divide temporal bone fractures with respect to the long axis of the petrous pyramid (**Fig. 3.5**): (1) **longitudinal fracture** (oriented along the petrous axis, parallel to the petrotympanic fissure), (2) **transverse fracture** (oriented perpendicular to the petrous axis), (3) **oblique** (oriented along the petrous axis, crossing the petrotympanic fissure), or (4) **mixed** fractures.

Oblique/longitudinal fractures represent 80% of all temporal bone fractures (~25% associated with facial nerve injury). Transverse fractures are less common; however, 40 to 50% are associated with facial palsy. Transverse fractures more commonly affect the otic capsule and cause sensorineural hearing loss.

Differential Diagnosis

There is a broad range of causes of trauma and associated injuries, some of which have been listed in preceding paragraphs.

◆ Evaluation

Physical Exam

Soft tissue injuries isolated to the auricle are evaluated with a focused otologic exam. Exposed cartilage, auricular hematoma, and auricular viability should be noted.

Patients with severe head or multisystem injury should receive standard trauma and resuscitation. Life-threatening injuries (e.g., cervical spine trauma, intracranial bleeding) must be treated emergently, and the patient should be stabilized. Complete head and neck, otologic, neurologic, and cranial nerve examinations are recommended.

When feasible, immediate or delayed facial paralysis should be assessed early and prior to intubation. The ear canal should be cleaned, and injuries to the EAC and tympanic membrane should be assessed. Presence of CSF leak should be ascertained. Hearing can be evaluated with a 512-Hz tuning fork, and the presence and type of nystagmus should be noted.

In patients with less severe injury, a more focused exam is feasible. Foreign body or penetrating injuries should be evaluated with an oto-microscope. In uncooperative children, there should be a low threshold for exam under anesthesia.

Imaging

High-resolution computed tomography (CT) of the temporal bone is recommended to assess for temporal bone fractures and integrity of the EAC, ossicles, otic capsule, middle and posterior cranial fossa plates, carotid canal, jugular bulb, and sigmoid sinus. Fractures involving the carotid canal and penetrating traumas (e.g., gunshot wounds) to the temporal bone require angiography to rule out vascular injury.

Other Tests

Pure tone and speech audiometry should be obtained when feasible. Electrophysiologic testing of the facial nerve (e.g., electroneuronography [ENoG] and electromyography [EMG]) may be helpful in terms of predicting recovery and guiding treatment decisions in patients with complete facial paralysis.

◆ Treatment Options

Soft Tissue Injuries

Auricular lacerations must be cleaned thoroughly. Cartilage may be reapproximated. When possible, soft tissue should be closed over exposed cartilage. If tissue is devitalized, wet-to-dry dressing coverage can be provided and surgical reconstruction planned in a delayed fashion. Antibiotics with adequate *Pseudomonas* coverage are prescribed.

Auricular hematomas must be drained to prevent a cauliflower deformity. To prevent reaccumulation, a pressure bolster is applied. Dental rolls or rolled Xeroform (Kendall Company, Mansfield, MA) are placed on the anterior and posterior surface of the auricle and secured with 2–0 nylon in a through-and-through mattress stitch for 5 days. Antibiotics are generally prescribed.

Auricular burns are treated topically with mafenide acetate. This sulfonamide-type medication can penetrate eschar and prevent cartilage loss. Infected cartilage is débrided. Treatment for frostbite injury is focused on rapid rewarming; avoidance of thawing and refreezing are critical for

auricular viability. Anti-inflammatories are also important for pain control and reducing inflammation. Prophylactic antibiotics are not routinely recommended. However, ongoing local wound care is often required as the injury demarcates.

Microsurgical revascularization and surgical reattachment of avulsions should be performed; however, there is a high failure rate, requiring delayed débridement and reconstruction. Most animal and human bites to the face or auricle can be thoroughly irrigated, débrided, sutured closed, and treated with antibiotics. Human bites have a higher risk of infection, and some clinicians advocate healing by secondary intention.

Tetanus vaccination, tetanus immunoglobulin, and rabies vaccination should be administered in the appropriate scenarios.

Penetrating Trauma to the External Auditory Canal and Middle Ear

Ear canal injuries should be cleaned under the microscope and stented if there is circumferential injury, to prevent stenosis. When stenting is performed with Oto-Wick (Medtronic Xomed, Inc., Jacksonville, FL) or Gelfoam (Pfizer Pharmaceuticals, New York, NY), antibiotic-and-steroid otic drops are prescribed.

Most traumatic tympanic membrane perforations can be observed for spontaneous resolution, because approximately 90% of small perforations resolves without treatment. Nonhealing tympanic membrane perforations can be treated with myringoplasty or tympanoplasty. Prophylactic antibiotic ear drops are prescribed for contaminated wounds. Dry ear precautions and avoidance of nose blowing are discussed.

Exploratory tympanotomy (myringotomy) and perilymphatic fistula repair is recommended if the injury is located in the posterosuperior quadrant of the tympanic membrane, severe vertigo is present, and stapes dislocation and perilymphatic fistula are suspected. Treatment of hearing loss and vertigo related to perilymphatic fistula may also require bed rest, systemic steroids, antiemetics, and short-term vestibulosuppressants.

Follow-up audiogram within three months of injury is recommended. Tympanoplasty and ossiculoplasty can be performed for persistent tympanic membrane perforation and ossicular discontinuity or fracture.

Temporal Bone Fracture and Other Injuries

Uncomplicated temporal bone fractures can be managed conservatively with wound care and follow-up audiograms. Otic-violating fractures or inner ear concussion causing sensorineural hearing loss should be treated with systemic steroids unless there is a medical contraindication. Vestibular dysfunction should be treated symptomatically with bed rest, antiemetics, and short-term vestibulosuppressants. Follow-up audiograms are recommended, and vestibular testing can be obtained for prolonged dizziness.

Immediate-onset facial paralysis after temporal bone fracture is thought to be from nerve transection, while incomplete facial palsy and delayed-onset facial paralysis may be due to nerve compression from edema or hematoma. High-dose steroids are recommended for facial nerve impairment. The prognosis for incomplete or delayed-onset facial paralysis is excellent, and

surgical intervention is rarely indicated. Immediate-onset paralysis has a worse prognosis. Surgical decompression or nerve repair is considered if there is > 90% degeneration on ENoG between 3 and 14 days after onset. EMG is performed if ENoG shows absent responses (phase cancellation can occur from simultaneous degeneration and regeneration). On EMG, no voluntary action potentials indicates a poor prognosis.

Gunshot wounds may involve widespread injury, carrying a high incidence of severe vascular injury and high mortality rate. Endovascular stenting may be necessary to prevent life-threatening hemorrhage from vascular injury. Most cases of CSF otorrhea or rhinorrhea resolve spontaneously with bed rest and stool softeners. Lumbar drain is considered if the CSF leak does not resolve in 5 to 7 days. Surgical intervention with removal of epithelial elements, closure of the ear canal, and auditory tube and mastoid obliteration with abdominal fat can be performed if lumbar drain fails to resolve the CSF leak. Antibiotic prophylaxis is controversial.

◆ Outcome and Follow-Up

Aural rehabilitation for permanent hearing loss can be offered. Treatment can include conventional hearing aid, bone-anchored hearing aid, contralateral routing of sound (CROS) hearing aids, and in extreme cases, cochlear implantation. Persistent vestibular complaints can be evaluated with vestibular testing and treated appropriately with particle repositioning for canalolithiasis and vestibular therapy for uncompensated vestibulopathy. In patients with facial paralysis, frequent wetting or lubrication of the eye is recommended; gold or platinum weight implants can be performed to assist in eye closure. Dynamic and static facial reanimation, nerve transfer (e.g., hypoglossal-to-facial nerve transposition, masseteric-to-facial nerve transposition), and free flap (e.g., gracilis free flap) reanimation can be discussed for poor nerve regeneration outcomes.

3.1.3 Acute Facial Paresis and Paralysis

◆ Key Features

- Facial paralysis in Bell's palsy occurs rapidly, usually within 24 to 48 hours.
- Treatment consists of high-dose oral steroids and antiviral medication.
- Eye care is essential and should not be overlooked.
- Spontaneous recovery to normal or near-normal function occurs in ~85%.

Bell's palsy, or acute idiopathic facial nerve palsy, accounts for 60 to 75% of all acute facial palsies. Bell's palsy is characterized by rapid-onset (24 to 48 hours) facial weakness that may progress to complete paralysis. A facial

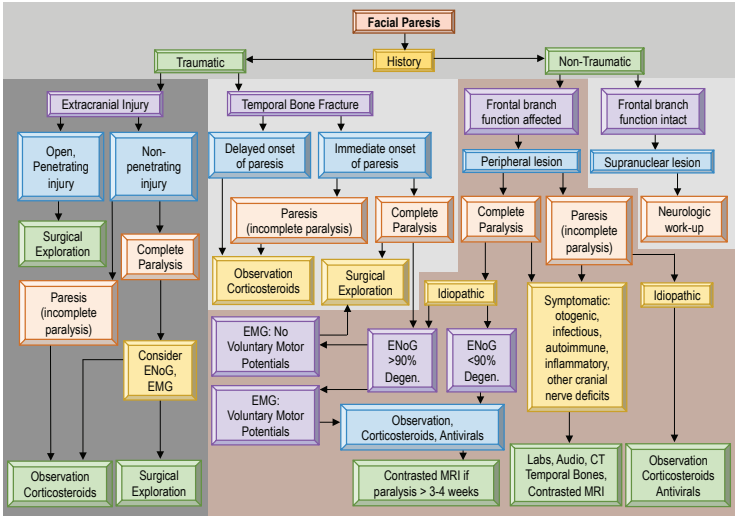


Fig. 3.6 Algorithm for the differential diagnosis of facial palsy. Many advocate surgical decompression only for cases of complete paralysis, and, although not specifically indicated on this algorithm, facial nerve decompression may be useful in cases of nontraumatic paralysis. (Modified with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:293.)

weakness that progresses slowly over weeks to months is suspicious for a neoplasm. Treatment for Bell's palsy consists of a prednisone taper (~ 1 mg/kg orally daily) plus an antiviral medication for at least one week. Eye lubrication is recommend if eye closure is compromised (see also **Chapter 9.2**). If there is little to no recovery of facial function after 2 to 3 months, then imaging is recommended to rule out neoplastic etiology.

◆ Epidemiology

The incidence of Bell's palsy is 20 to 30 cases per 100,000 people per year. It accounts for almost 75% of all unilateral facial palsy; 40,000 cases occur in the United States each year. Median age of onset is 40 years, but the incidence is highest over age 70. Men and women, and right and left sides, are affected equally. Pregnant women and patients with diabetes and hypertension are at increased risk.

Table 3.3 The House–Brackmann Facial Nerve Grading Scale

Grade	Facial nerve findings
I. Normal	Normal
II. Mild dysfunction	Slight weakness on close inspection
III. Moderate dysfunction	Appears normal at rest; weakness with effort but still full eye closure
IV. Moderately severe dysfunction	Weakness visible at rest; incomplete eye closure
V. Severe dysfunction	Only barely perceptible motion
VI. Total paralysis	Complete paralysis

Data from House JW, Brackmann DE. Facial nerve grading system. *Otolaryngol Head Neck Surg* 1985;93(2):146–147.

◆ **Clinical**

Signs and Symptoms

Patients usually present with rapid-onset (24–48 hours) facial nerve weakness that may progress to complete paralysis. Patients often report pain and numbness around the ear, hyperacusis, and dysgeusia; 70% of patients will have a preceding viral illness.

Differential Diagnosis

Idiopathic facial nerve palsy is a diagnosis of exclusion (**Fig. 3.6**). Herpes zoster oticus (Ramsay-Hunt) is characterized by severe otalgia and vesicular lesions involving the ear and accounts for 10 to 15% of acute facial palsies. Melkersson-Rosenthal’s syndrome consists of recurrent bouts of unilateral facial palsy in association with facial edema and a fissured tongue. Facial palsy occurs in 11% of patients with Lyme disease and may be bilateral. Acute and chronic otitis media with or without cholesteatoma can also cause acute facial palsy, as can necrotizing or malignant otitis externa. Neoplasms need to be ruled out if there is no symptomatic improvement after 2 to 3 months or if the palsy is characterized by slow onset or relapse. Temporal bone trauma can also result in acute facial paralysis.

◆ **Evaluation**

History

The most important information is the history. Time of onset and history of previous episodes are paramount. Hearing loss and vertigo are also important to note. Patients with herpes zoster oticus are likely to experience involvement of the vestibulocochlear nerve (CN VIII).

Physical Exam

A complete head and neck examination is required. Determination is made whether facial nerve function is weak or completely absent. The House–Brackmann scale for facial function is often used for charting and physician communication (**Table 3.3**). If the upper face is spared, then a central etiology

is suspected. The eardrum is carefully inspected to rule out acute or chronic middle ear disease, cholesteatoma, or a temporal bone neoplasm, either benign or malignant. The skin is evaluated for signs of vesicular lesions. Vesicular lesions may be periauricular, auricular, in the ear canal, or even on the palate. They are generally tender and present in various stages of healing. Palpation of the neck and parotid gland is crucial in ruling out an extratemporal process. The remaining cranial nerves are also examined, looking for evidence for polyneuropathy.

Ancillary Tests

An audiogram is recommended when there are otologic complaints or prolonged facial paralysis. Asymmetry in hearing may suggest cochlear, retrocochlear, or cerebellopontine angle process contributing to facial weakness.

Imaging

Imaging is not routinely obtained at presentation if the history is consistent with acute idiopathic palsy. If the history is suspicious for neoplasm or otologic symptoms are present, then a magnetic resonance imaging (MRI) scan of the head with and without gadolinium enhancement is obtained to visualize the entire course of the facial nerve. If there is any enhancement along the intratemporal course of the nerve, a noncontrast fine-cut temporal bone computed tomography (CT) scan will help to delineate the facial (fallopian) canal looking for dilation or erosion. If the palsy shows no signs of improvement within 3 months, imaging is recommended.

Labs

Labs are not routinely ordered. If the clinical picture does not follow that of an idiopathic palsy or there are risk factors for other disorders, then labs are ordered as indicated. Systemic disorders causing facial weakness include autoimmune syndromes, Wegener's granulomatosis, sarcoidosis, Lyme disease, and syphilis.

Electrical Testing

Electrical testing continues to be controversial. If a patient is evaluated within the first 2 weeks of the onset, an electroneuronography (ENoG) can be obtained. It is useful when performed within 3 to 14 days of onset of total paralysis and when the contralateral face is unaffected. A stimulating surface electrode is placed at the stylomastoid foramen, and a recording surface electrode is placed at the nasolabial fold. A compound muscle action potential is then recorded and compared with that recorded from the unaffected side. Changes in amplitude are calculated and interpreted as a percentage of nerve fiber dysfunction. If serial ENoGs are performed starting after day 3, and the amplitude on the affected side indicates > 90% loss of function, then surgical decompression may be an option. Electromyography (EMG) is also recommended in this circumstance, and surgical candidates are expected to demonstrate no voluntary action potentials within 2 weeks of paralysis.

Pathology

Similar to idiopathic SNHL, idiopathic facial nerve palsy as a symptom is likely secondary to multiple insults. Theories focus on virally mediated

inflammation followed by ischemia, neural blockade, and degeneration. The inflammation leads to physical compression as the facial nerve enters the fundus of the temporal bone and forms the labyrinthine segment. This region is known as the meatal foramen and is the narrowest portion of the facial canal, forming a bottleneck area. Nerve conduction studies on patients undergoing decompression surgery for Bell's palsy have consistently demonstrated decreased electrical responses distal to the meatal foramen. Histopathologic studies obtained from autopsy of patients who died shortly after onset of Bell's palsy have been reported. The findings are similar though not uniform. Inflammatory neuritis consistent with a viral infection is most often reported, but intraneural vascular congestion and hemorrhage have also been identified. Viral studies have shown evidence implicating herpes simplex virus (HSV) as the offending agent. Herpes virus has been harvested from nasal and oral secretions of patients experiencing an acute facial palsy. Animal models have also been used to demonstrate acute facial paralysis after intraoral and periauricular inoculation with HSV. All of these studies are persuasive but as yet not conclusive.

◆ Treatment Options

Medical

Management of Bell's palsy consists of high-dose oral steroids (e.g., prednisone ~1 mg/kg start dose) with a 2- to 3-week taper and a concurrent 7 to 10 days of an oral antiviral (e.g., valacyclovir 500 mg orally three times daily). The natural history of the disorder suggests that almost 85% of patients will recover normal or near-normal function without treatment, but there is no test to determine outcomes at the time of presentation. Therefore, all patients are treated with this medical regimen.

Patients must also be evaluated for the need for eye care. If paralysis is complete and facial tone is poor, there will be corneal exposure throughout the day and night. Patients receive frequent doses of artificial tears to that eye and use a nighttime lubricant to protect the cornea while they sleep. They can also wear a clear moisture chamber over the eye.

Surgical

Surgery for these patients remains controversial. Centers that see large volumes of these patients have developed algorithms that include serial ENoG testing in conjunction with EMG. If patients present within 2 weeks of onset of paralysis, and reach > 90% facial nerve degeneration without evidence of EMG action potentials, they are offered middle fossa facial nerve decompression. The middle fossa route enables complete decompression of the labyrinthine portion of the facial nerve, including the meatal foramen, without sacrificing hearing. Small series have shown that patients choosing decompression have a 91% chance of good recovery (House–Brackmann I or II). Patients who refuse surgery have a 58% chance of poor recovery (House–Brackmann III or IV).

Patients who experience poor recovery often require facial enhancement or reanimation procedures. Tarsorrhaphy, spring, or gold weight should be recommended to prevent or treat corneal abrasions. Refer to **Chapter 9.2**

for more details. Botulinum toxin A has been used to help with bothersome blepharospasm due to synkinesis.

◆ Outcome and Follow-Up

Patients are followed longitudinally to assess final recovery. Final recovery from a facial nerve injury may not be seen until 18 to 24 months after insult. According to Peiterson's article charting the natural history of untreated Bell's palsy, 71% of patients will recover completely, and an additional 13% will achieve near normal recovery (House–Brackmann I or II). Anyone with poor recovery or at risk for corneal abrasion should be considered for surgical reanimation or surgical management of the paralyzed eye. Patients with herpes zoster oticus are at a much higher risk (almost 50%) to experience poor recovery (House–Brackmann III or worse). Patients who recover but then experience a recurrence of symptoms should be evaluated for a neoplasm, Melkersson-Rosenthal's syndrome, or inflammatory/autoimmune conditions.

3.1.4 Ear Foreign Bodies

◆ Key Features

- Ear foreign bodies are commonly found in young children.
- The injury may be acute or chronic, with associated infection.
- Foreign bodies often consist of insects, small batteries, food, beads, pencil erasers, and other inert objects.
- Injury to the tympanic membrane, ossicles, facial nerve, or inner ear must be considered.

Foreign bodies in the external auditory canal are a common problem, especially in pediatric patients. The otolaryngologist often becomes involved after unsuccessful attempts at removal by the patient, parent, emergency room providers, and/or pediatricians. With children, one should have a low threshold for removal under anesthesia in the operating room.

◆ Epidemiology

An ear foreign body is most commonly seen in children under 12 years of age.

◆ Clinical

Signs and Symptoms

An acute foreign body may cause otalgia due to pressure on thin skin of the medial ear canal. Insects are extremely bothersome if alive. Obstruction may cause conductive hearing loss. Chronic foreign bodies, especially organic material such as food, may present with foul otorrhea or purulence.

Pronounced hearing loss, vertigo, or facial weakness suggests injury to the middle ear (see **Chapter 3.1.2**).

Differential Diagnosis

The differential diagnosis includes a foreign body, otitis externa, otitis media, middle ear trauma, or a tumor.

◆ Evaluation

History

When taking the patient's history, you should inquire about the time course and possible nature of foreign body, if unclear. The previous level of hearing is important, as is a history of known tympanic membrane perforation. Injury involving an only-hearing ear mandates careful and experienced management.

Physical Exam

The examination should include a full head and neck exam to evaluate for other foreign bodies. Document facial nerve function and perform a 512-Hz tuning fork exam if the patient is old enough and cooperative. The ear should be examined with the otomicroscope. Any purulence or drainage is suctioned with #3 or #5 Frazier tip suction to enable visualization of the foreign body, canal skin, and tympanic membrane, if possible.

Imaging

High-resolution CT scanning of the temporal bone is the most useful imaging study if there is concern for a middle or inner ear injury.

Other Tests

Perform an audiologic assessment if there is any concern of a middle or inner ear injury.

◆ Treatment Options

Most foreign bodies do not constitute an emergency and can be removed within a reasonable time period (i.e., days). Exceptions requiring prompt treatment include batteries (which can cause tissue necrosis due to alkaline or acid leakage), insects, or objects associated with additional trauma to the middle ear or temporal bone. In the adult or cooperative child, one may be able to remove a foreign body easily in the office. If a child is at all uncooperative, it is preferable to proceed with removal under anesthesia in the operating room in an elective setting. Live insects are first treated by instillation of 1% lidocaine to the canal, which promptly drowns/paralyzes the insect and anesthetizes the canal skin. The most useful instrument is a blunt otologic right angle hook, which may be positioned medial to the foreign body, enabling removal. This is especially helpful for smooth objects, such as beads. An object that can be grasped and is intact may be amenable to the use of alligator forceps. Longstanding decaying organic material associated with infection requires suctioning. Gravel or sand against an intact

tympanic membrane is best removed with irrigation. Following removal of a foreign body, careful inspection of the canal skin and tympanic membrane is important. Topical antibiotic and steroid drops for a few days are often helpful in treating or preventing local infection. Oral antibiotics are added for cellulitis, chondritis, and otitis media, as well as for the diabetic or immunosuppressed patient.

The presence of canal skin edema may require the placement of an ear wick (e.g., Meroce[®]l Pope Oto-Wick, Medtronic, Inc., Minneapolis, MN). Necrotic tissue secondary to acid or alkaline battery leakage or thermal burns from slag foreign bodies (e.g., due to welding) are managed with serial débridement and antibiotic steroid drops; wicks or stenting are required for a circumferential canal injury. Management of injuries to the tympanic membrane and the middle ear, as well as temporal bone penetrating trauma, are discussed in **Chapter 3.1.2**.

◆ **Outcome and Follow-Up**

Uncomplicated removal of a foreign body usually does not require follow-up. If there is any associated infection, follow-up within a few weeks to document resolution following topical therapy is routine.

